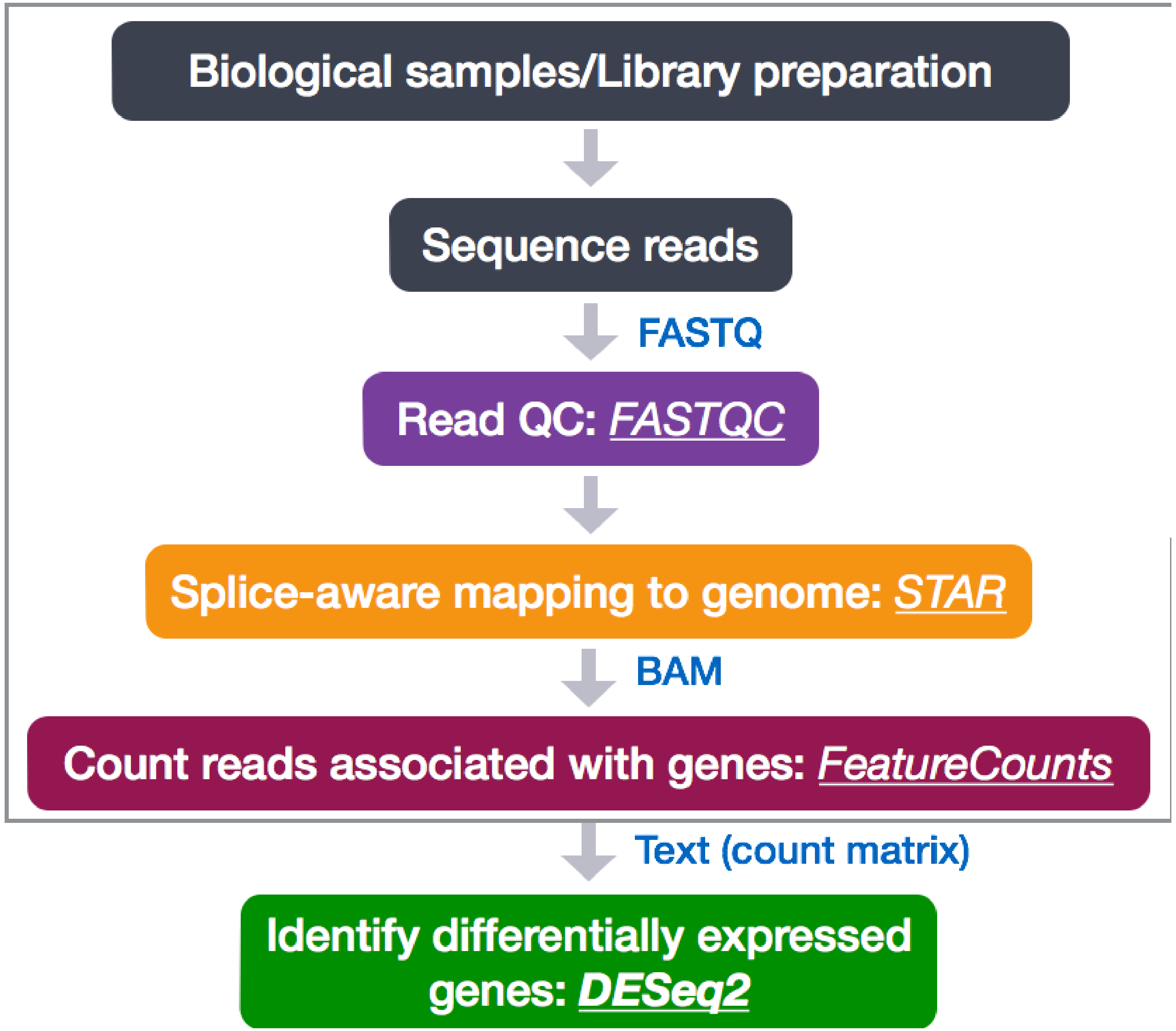
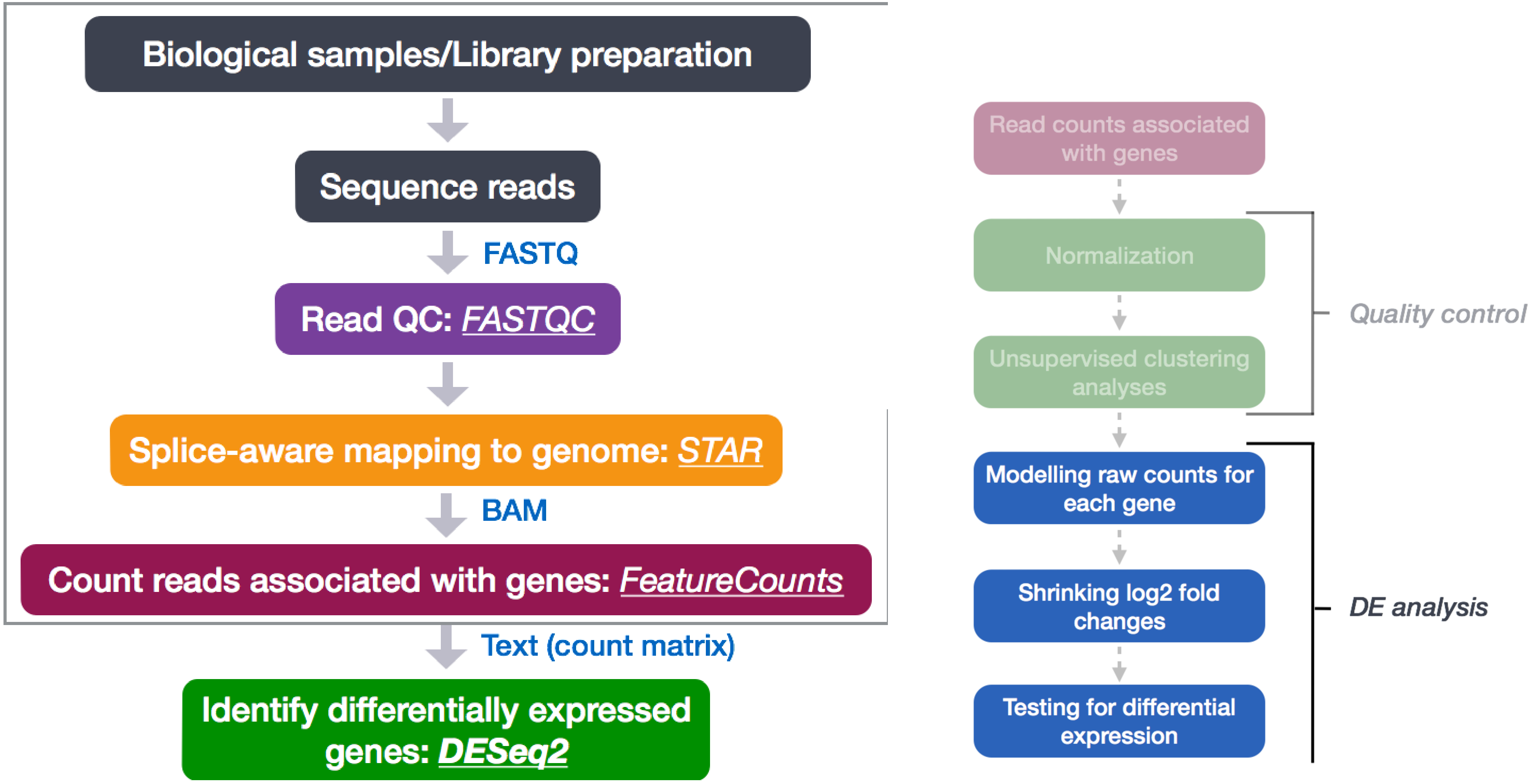
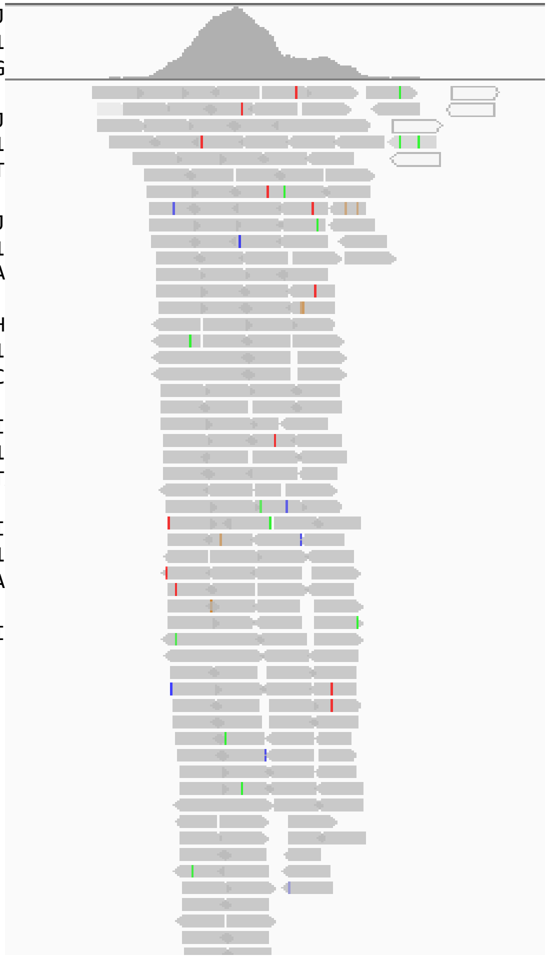

Biological samples/Library preparation
Sequence reads
FASTQ
Read QC: FASTQC
Splice-aware mapping to genome: STAR
BAM
Count reads associated with genes: FeatureCounts
Text (count matrix)
Identify differentially expressed genes: DESeq2



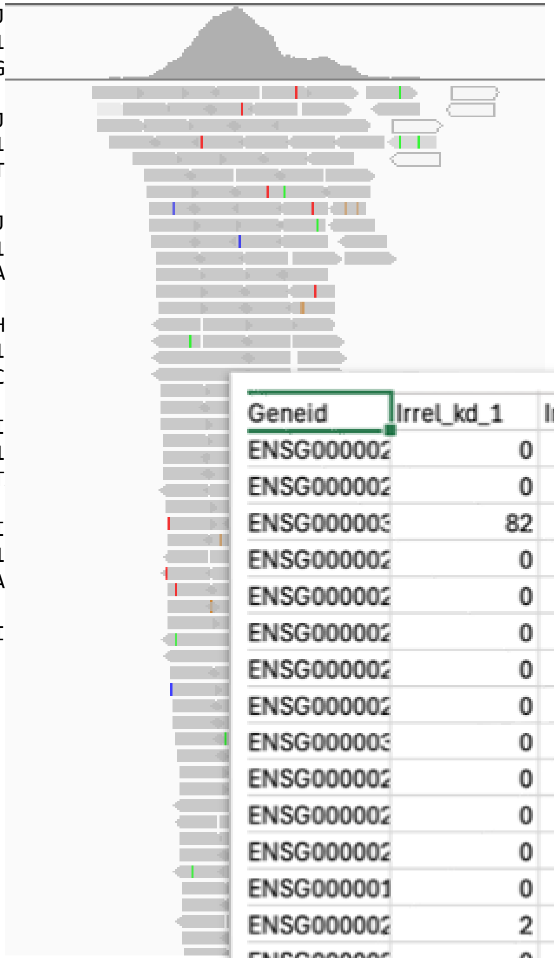
Biological samples/Library preparation
Sequence reads
FASTQ
Read QC: FASTQC
Splice-aware mapping to genome: STAR
BAM
Count reads associated with genes: FeatureCounts
Text (count matrix)
Identify differentially expressed genes: DESeq2
Read counts associated with genes
Normalization
Unsupervised clustering analyses
Quality control
Modelling raw counts for each gene
Shrinking log2 fold changes
Testing for differential expression
DE analysis

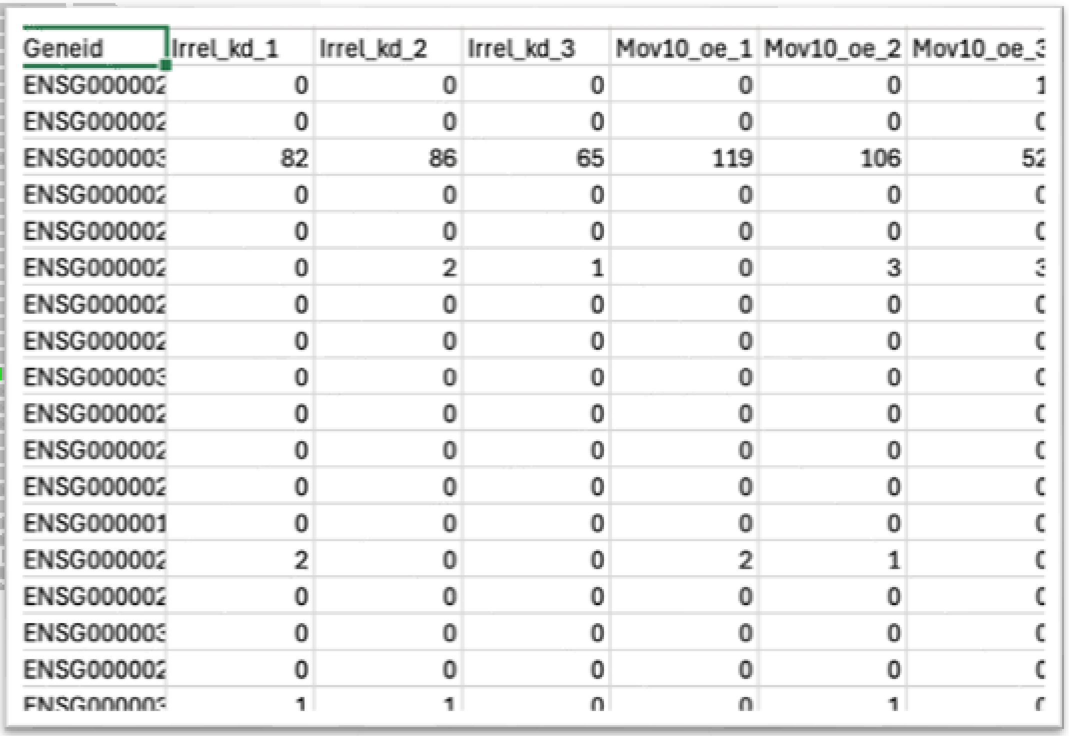
```
@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCCAACCGTACAGGTGGGTCTCTCCCTCATGGA
+
CCCFFFFFHHHHHGHIIJIJJJIEHGIJJCGGJAFHEIJDHGIFGGE<
@SRR948304.5160 UNC14-SN744:253:D135LACXX:5:1101:4625:4251
CCACGAAGTCAAGATCGCCGACAAGGCCTTCCTGATGAAGCAGAAGTT
+
BCCFFFFFHGHHHJJJJJJJIJGJJJJJJJJJJFIJJGIJJIJJJIGI
@SRR948304.5196 UNC14-SN744:253:D135LACXX:5:1101:5864:4425
GGAACGCCTACTGGTACTATCTGAACATGGACTACTCCTTCTTCCTGG
+
CCCFFFFFFHHHHJGHIIJJJJJJJJIJJJIJIIJJJJJJJJJJJJJI
@SRR948304.5200 UNC14-SN744:253:D135LACXX:5:1101:6109:4282
CAGGAAGAAGGAGTAGTCCATGTTCAGATAGTACCAGTAGGCGTTCCA
+
@@CFFBDDHHHBHIHIFGHHIGFFHHIJJJJFIIJIJGDFHIIFGGIF
@SRR948304.5215 UNC14-SN744:253:D135LACXX:5:1101:6464:4443
CCTTCTTCAACGACTACTACACCAAGTACTTCAAGTTCTGAGCGTCGT
+
@@@FFFFFHGDHFIEHGIJIFEHIHH?EFGGIGGG:BFGG4DFE)?D6
@SRR948304.5247 UNC14-SN744:253:D135LACXX:5:1101:8102:4331
CTACAGCTTCCGCAAGAACTACTACGAGATGGAGACCTACGCCAACTA
+
BB@FFFFFHHHHFIJIJJJIJIJIJJIGGGIIGGGIIJIJGIIGIII#
@SRR948304.5250 UNC14-SN744:253:D135LACXX:5:1101:8320:4324
CCTGGTTGAACTCGTAGATATTCTCGCGCAGAATGAACTTGTCCTCGG
+
CCCFFFFFHHHHHJHGHIJIJJJJJJJJJJJIIIJJIJJJJJJIJJJE
```

```
@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCCAACCGTACAGGTGGGTCTCTCCCTCATGGA
+
CCCFFFFFHHHHHGHIIJIJ
@SRR948304.5160 UNC1
CCACGAAGTCAAGATCGCCG
+
BCCFFFFFHGHHHJJJJJJJ
@SRR948304.5196 UNC1
GGAACGCCTACTGGTACTAT
+
CCCFFFFFFFHHHHJGHIIJJ
@SRR948304.5200 UNC1
CAGGAAGAAGGAGTAGTCCA
+
@@CFFBDDHHHBHIHIFGHH
@SRR948304.5215 UNC1
CCTTCTTCAACGACTACTAC
+
@@@FFFFFHGDHFIEHGIJI
@SRR948304.5247 UNC1
CTACAGCTTCCGCAAGAACT
+
BB@FFFFFHHHHFIJIJJJI
@SRR948304.5250 UNC1
CCTGGTTGAACTCGTAGATA
+
CCCFFFFFFHHHHHJHGHIJI
```



```
@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCCAACCGTACAGGTGGGTCTCTCCCTCATGGA
+
CCCFFFFFHHHHHGHIIJIJ
@SRR948304.5160 UNC1
CCACGAAGTCAAGATCGCCG
+
BCCFFFFFHGHHHJJJJJJJ
@SRR948304.5196 UNC1
GGAACGCCTACTGGTACTAT
+
CCCFFFFFFHHHHJGHIIJJ
@SRR948304.5200 UNC1
CAGGAAGAAGGAGTAGTCCA
+
@@CFFBDDHHHBHIHIFGHH
@SRR948304.5215 UNC1
CCTTCTTCAACGACTACTAC
+
@@@FFFFFHGDHFIEHGIJI
@SRR948304.5247 UNC1
CTACAGCTTCCGCAAGAACT
+
BB@FFFFFHHHFIJIJJJI
@SRR948304.5250 UNC1
CCTGGTTGAACTCGTAGATA
+
CCCFFFFFHHHHHJHGHIJI
```

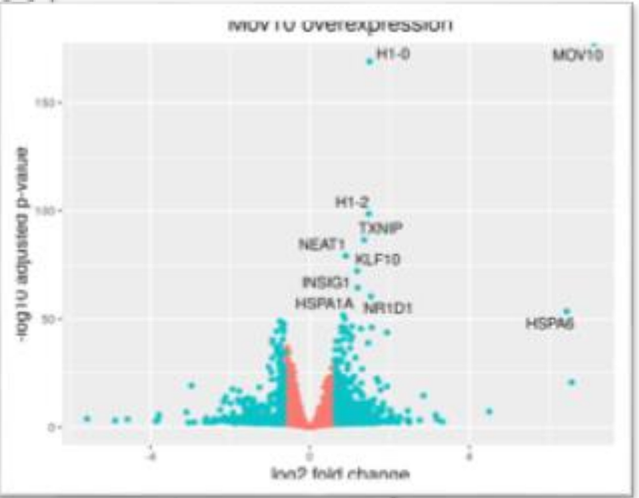




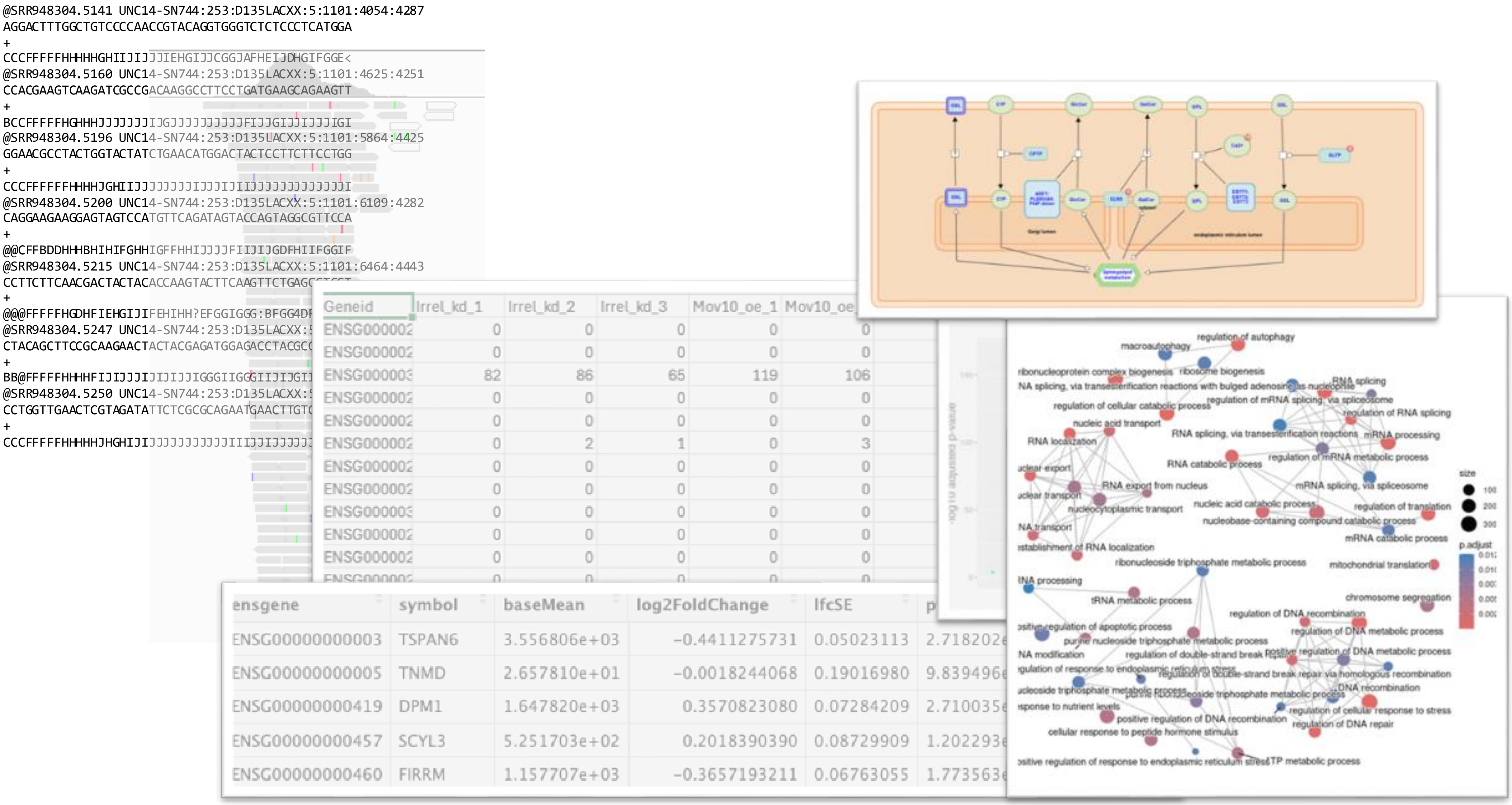
Geneid	Irrel_kd_1	Irrel_kd_2	Irrel_kd_3	Mov10_oe_1	Mov10_oe_2	Mov10_oe_3
ENSG000002	0	0	0	0	0	1
ENSG000002	0	0	0	0	0	0
ENSG000003	82	86	65	119	106	52
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	2	1	0	3	3
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000001	0	0	0	0	0	0
ENSG000002	2	0	0	2	1	0
ENSG000002	0	0	0	0	0	0
ENSG000003	0	0	0	0	0	0
ENSG000002	0	0	0	0	0	0
ENSG000003	1	1	0	0	1	0

```
@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCCAACCGTACAGGTGGGTCTCTCCCTCATGGA
+
CCCFFFFFHHHHHGHIIJIJJJJIEHGIJJCGGJAFHEIJDHGIFGGE<
@SRR948304.5160 UNC14-SN744:253:D135LACXX:5:1101:4625:4251
CCACGAAGTCAAGATCGCCGACAAGGCCTTCCTGATGAAGCAGAAGTT
+
BCCFFFFFHGHHHJJJJJJJIJGJJJJJJJJJJFIJJGIJJIJJJJIGI
@SRR948304.5196 UNC14-SN744:253:D135LACXX:5:1101:5864:4425
GGAACGCCTACTGGTACTATCTGAACATGGACTACTCCTTCTTCCTGG
+
CCCFFFFFFHHHHJGHIIJJJJJJJIJJJIJIIJJJJJJJJJJJJJI
@SRR948304.5200 UNC14-SN744:253:D135LACXX:5:1101:6109:4282
CAGGAAGAAGGAGTAGTCCATGTTCAGATAGTACCAGTAGGCGTTCCA
+
@@CFFBDDHHHBHIHIFGHHIGFFHHIJJJJFIIJIJGDFHIIFGGIF
@SRR948304.5215 UNC14-SN744:253:D135LACXX:5:1101:6464:4443
CCTTCTTCAACGACTACTACACCAAGTACTTCAAGTTCTGAG
+
@@@FFFFFHGDHFIEHGIJIFEHIHH?EFGGIGGG:BFGG4DF
@SRR948304.5247 UNC14-SN744:253:D135LACXX:5
CTACAGCTTCCGCAAGAACTACTACGAGATGGAGACCTACGC
+
BB@FFFFFHHHHFIJIJJJIJIJIJJIGGGIIGGGIIJIJGI
@SRR948304.5250 UNC14-SN744:253:D135LACXX:5
CCTGGTTGAACTCGTAGATATTCTCGCGCAGAATGAACTTGT
+
CCCFFFFFFHHHHHJHGHIJIJJJJJJJJJJJIIIJJIJJJJJ
```

Geneid	Irrel_kd_1	Irrel_kd_2	Irrel_kd_3	Mov10_oe_1	Mov10_oe_2	Mov10_oe_3
ENSG000002	0	0	0	0	0	
ENSG000002	0	0	0	0	0	
ENSG000003	82	86	65	119	106	
ENSG000002	0	0	0	0	0	
ENSG000002	0	0	0	0	0	
ENSG000002	0	2	1	0	3	
ENSG000002	0	0	0	0	0	
ENSG000002	0	0	0	0	0	
ENSG000003	0	0	0	0	0	
ENSG000002	0	0	0	0	0	
ENSG000002	0	0	0	0	0	



ensgene	symbol	baseMean	log2FoldChange	lfcSE	p	
ENSG00000000003	TSPAN6	3.556806e+03	-0.4411275731	0.05023113	2.718202e-19	1.974760e-17
ENSG00000000005	TNMD	2.657810e+01	-0.0018244068	0.19016980	9.839496e-01	9.916255e-01
ENSG00000000419	DPM1	1.647820e+03	0.3570823080	0.07284209	2.710035e-07	2.804880e-06
ENSG00000000457	SCYL3	5.251703e+02	0.2018390390	0.08729909	1.202293e-02	3.882876e-02
ENSG00000000460	FIRRM	1.157707e+03	-0.3657193211	0.06763055	1.773563e-08	2.288515e-07

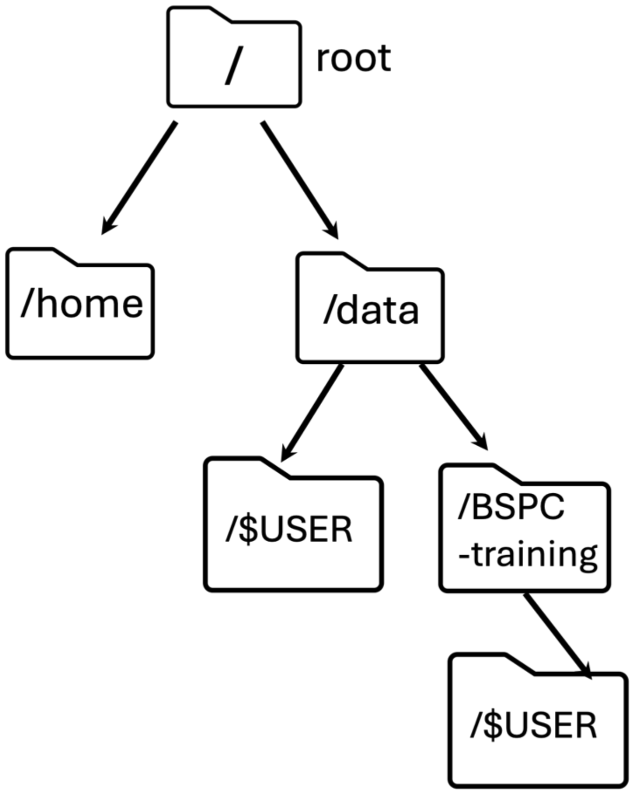


@SRR948304.5141 UNC14-SN744:253:D135LACXX:5:1101:4054:4287
AGGACTTTGGCTGTCCCCAACCGTACAGGTGGGTCTCTCCCTCATGGA
+
CCCFFFFFHHHHHGHIIJIJJJIEHGIJJCGGJAFHEIJDHGIFGGE<
@SRR948304.5160 UNC14-SN744:253:D135LACXX:5:1101:4625:4251
CCACGAAGTCAAGATCGCCGACAAGGCCTTCCTGATGAAGCAGAAGTT
+
BCCFFFFFHGHHHJJJJJJIJGJJJJJJJJJJFIJJGIJJIJJJJIGI
@SRR948304.5196 UNC14-SN744:253:D135LACXX:5:1101:5864:4425
GGAACGCCTACTGGTACTATCTGAACATGGACTACTCCTTCTTCCTGG
+
CCCFFFFFFHHHHJGHIIJJJJJJJJIJJJJIJIIJJJJJJJJJJJJI
@SRR948304.5200 UNC14-SN744:253:D135LACXX:5:1101:6109:4282
CAGGAAGAAGGAGTAGTCCATGTTCAGATAGTACCAGTAGGCGTTCCA
+
@@CFFBDDHHHBHIHIFGHHIGFFHHIJJJJFIIJIJGDFHIIFGGIF
@SRR948304.5215 UNC14-SN744:253:D135LACXX:5:1101:6464:4443
CCTTCTTCAACGACTACTACACCAAGTACTTCAAGTTCTGAG
+
@SRR948304.5247 UNC14-SN744:253:D135LACXX:5
+
@SRR948304.5250 UNC14-SN744:253:D135LACXX:5
+
Geneid
Irrel_kd_1
Irrel_kd_2
Irrel_kd_3
Mov10_oe_1
ensgene
symbol
baseMean
log2FoldChange
lfcSE
ENSG00000000003 TSPAN6 3.556806e+03 -0.4411275731 0.05023113
ENSG00000000005 TNMD 2.657810e+01 -0.0018244068 0.19016980
ENSG00000000419 DPM1 1.647820e+03 0.3570823080 0.07284209
ENSG00000000457 SCYL3 5.251703e+02 0.2018390390 0.08729909
ENSG00000000460 FIRRM 1.157707e+03 -0.3657193211 0.06763055
regulation of autophagy
macroautophagy
ribonucleoprotein complex biogenesis
ribosome biogenesis
RNA splicing
regulation of cellular catabolic process
regulation of mRNA splicing, via spliceosome
regulation of RNA splicing
nucleic acid transport
RNA localization
RNA splicing, via transesterification reactions
mRNA processing
RNA catabolic process
regulation of mRNA metabolic process
RNA export from nucleus
mRNA splicing, via spliceosome
nucleocytoplasmic transport
nucleic acid catabolic process
regulation of translation
nucleobase-containing compound catabolic process
mRNA catabolic process
ribonucleoside triphosphate metabolic process
mitochondrial translation
tRNA metabolic process
chromosome segregation
regulation of DNA recombination
regulation of DNA metabolic process
purine nucleoside triphosphate metabolic process
positive regulation of DNA metabolic process
DNA recombination
regulation of cellular response to stress
positive regulation of DNA recombination
regulation of DNA repair
cellular response to peptide hormone stimulus
ATP metabolic process
size
p.adjust

Quick Week-by-Week Summary!

Week 1

- Introduction to Shell and the File System
- Wildcards and Shortcuts
- Working with Files
- Searching in Files
- Data Organization



What commands could you use to navigate from THIS directory to THAT directory?

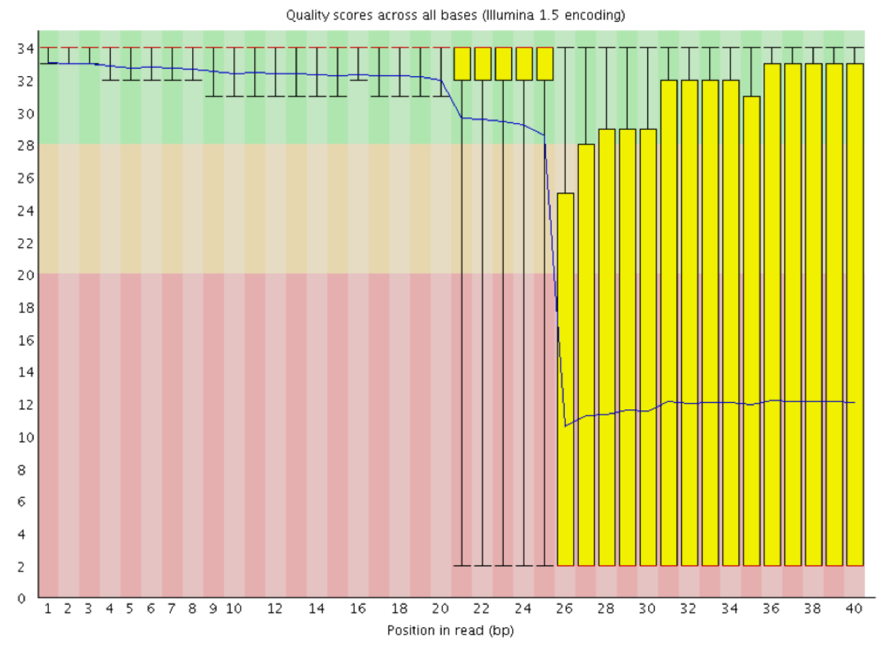
If you are editing a document in Vim – how do you save your current file AND exit at the same time?

grep -B 1 -A 2 NNNNNNNNNN Mov10_oe_2.subset.fq > bad_reads.txt

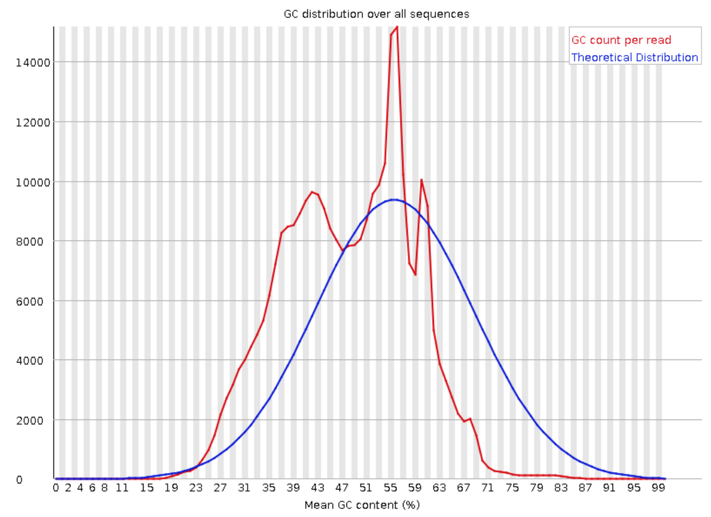
What are –B and –A? Will this overwrite anything already in bad_reads.txt?

Week 2

- [Running FASTQC Interactively]
- [Running FASTQC Using Job Submission]
- [FASTQC Assessment]

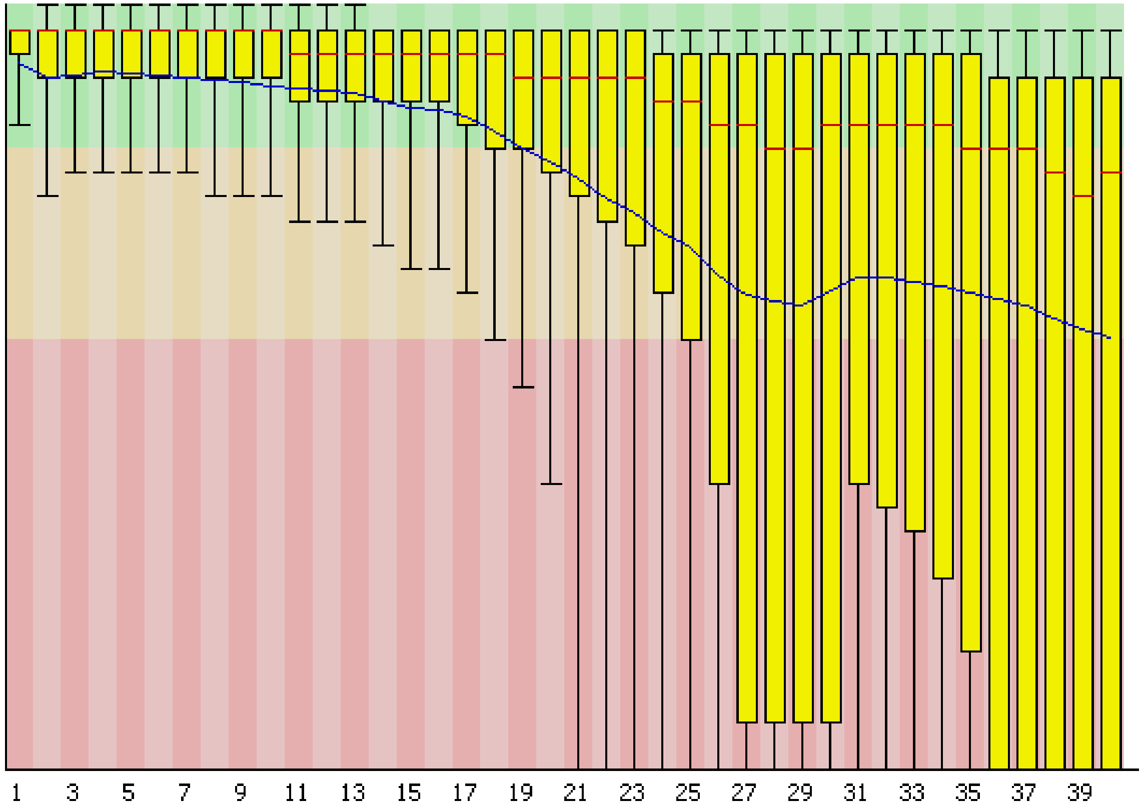


Per sequence GC content



What does this command do and how would I specify 8G of memory for it?

```
sinteractive --cpus-per-task=4
--mem=8g
```



How should I label the axes of this plot? Is this a worrying pattern?

Uh oh – Biowulf can't find the command `fastqc` even though I'm on an interactive node – what do I need to do??

Week 3

- Reference Genomes and Genome Indices
- Mapping with STAR and Viewing Results
- MultiQC for organizing results
- SAM/BAM format for mapping results

What ingredients do you need to create a genome index? What is a genome index used for?

What file format is this?

```
HWI-ST330:304:H045HADXX:2:2212:18192:97569 4 * 0 0 * * 0
CCAGATTTGCGATCTTCATTCCATACAGCATGGAGGAGAAGCCAGGGGCAGGGGTCCCATAGCTGGGCTTGAAGTCGTGGTTGTA
AATCGTCCGCAGCCG
=@@BDFFFF<FF8BEFE<FHGCFHE?@<++1:8:8CB13BBD;008(('70/';54?9=.7.;@?@==35>B@####################
####### NH:i:0 HI:i:0 AS:i:57 nM:i:3 uT:A:1
```

What are some parts of this data format?

Week 4

- Counting RNAseq Reads
- Experimental Design

Each column is a sample

Each row is a gene

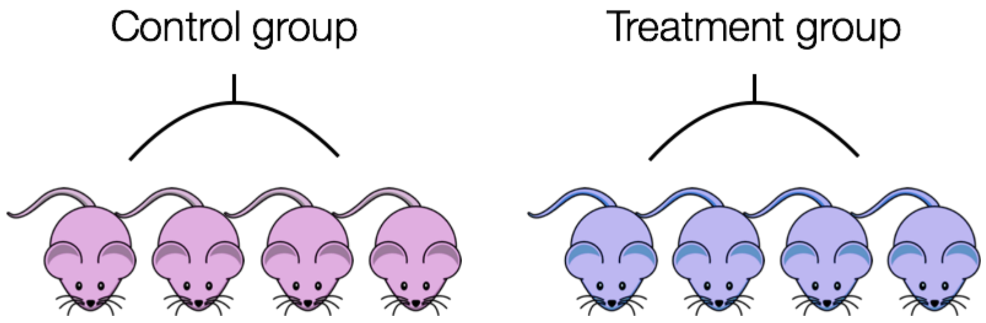
GENE ID	KD.2	KD.3	OE.1	OE.2	OE.3	IR.1	IR.2	IR.3
1/2-SBSRNA4	57	41	64	55	38	45	31	39
A1BG	71	40	100	81	41	77	58	40
A1BG-AS1	256	177	220	189	107	213	172	126
A1CF	0	1	1	0	0	0	0	0
A2LD1	146	81	138	125	52	91	80	50
A2M	10	9	2	5	2	9	8	4
A2ML1	3	2	6	5	2	2	1	0
A2MP1	0	0	2	1	3	0	2	1
A4GALT	56	37	107	118	65	49	52	37
A4GNT	0	0	0	0	1	0	0	0
AA06	0	0	0	0	0	0	0	0
AAA1	0	0	1	0	0	0	0	0
AAAS	2288	1363	1753	1727	835	1672	1389	1121
AACS	1586	923	951	967	484	938	771	635
AACSP1	1	1	3	0	1	1	1	3
AADAC	0	0	0	0	0	0	0	0
AADACL2	0	0	0	0	0	0	0	0
AADACL3	0	0	0	0	0	0	0	0
AADACL4	0	0	1	1	0	0	0	0
AADAT	856	539	593	576	359	567	521	416
AAGAB	4648	2550	2648	2356	1481	3265	2790	2118
AAK1	2310	1384	1869	1602	980	1675	1614	1108
AAMP	5198	3081	3179	3137	1721	4061	3304	2623
AANAT	7	7	12	12	4	6	2	7
AARS	5570	3323	4782	4580	2473	3953	3339	2666

Week 4

What do we input to featureCounts?

Anyone remember what this did for us?

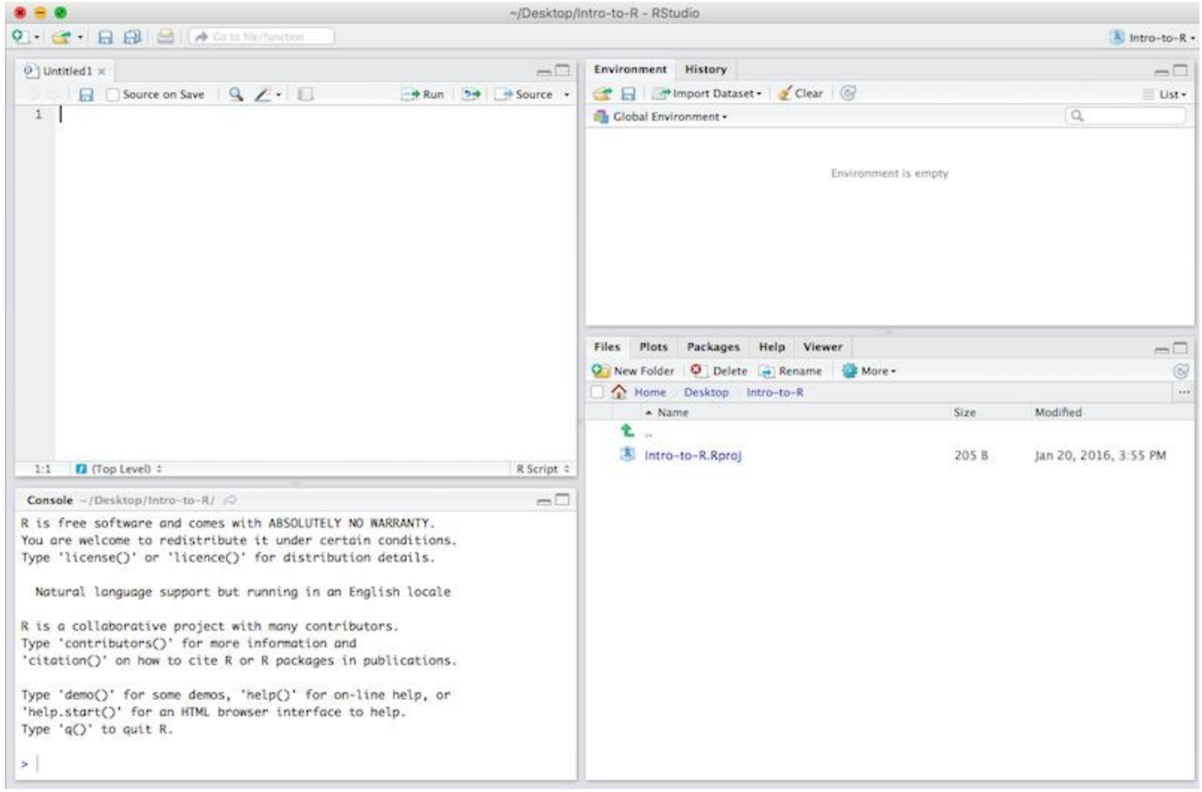
```
:%s/\/data\/Bspc-training\/shared\/rnaseq_jan2025\/results_for_counting\///g
```



How could we avoid confounding by sex (color)?

Week 5

- [Automating the RNAseq workflow]()
- [Intro to R and RStudio]()



Week 5

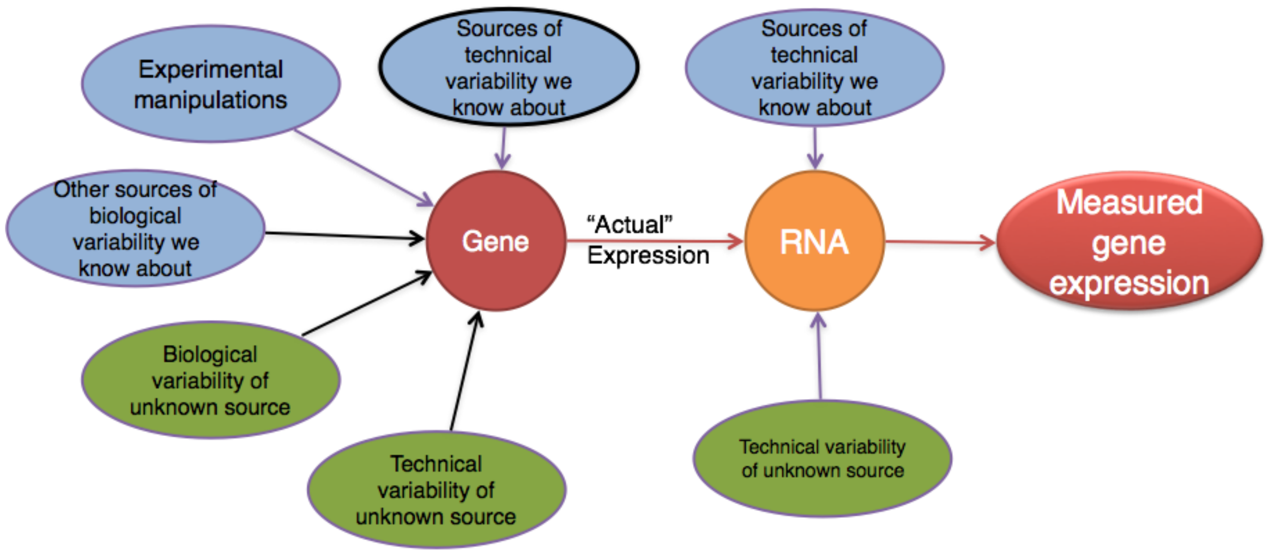
What does ` fq=${1} ` do in our automation script?

Where does the value for SLURM_CPUS_PER_TASK come from?

What are some ways we can check if a TSV was successfully imported into Rstudio?

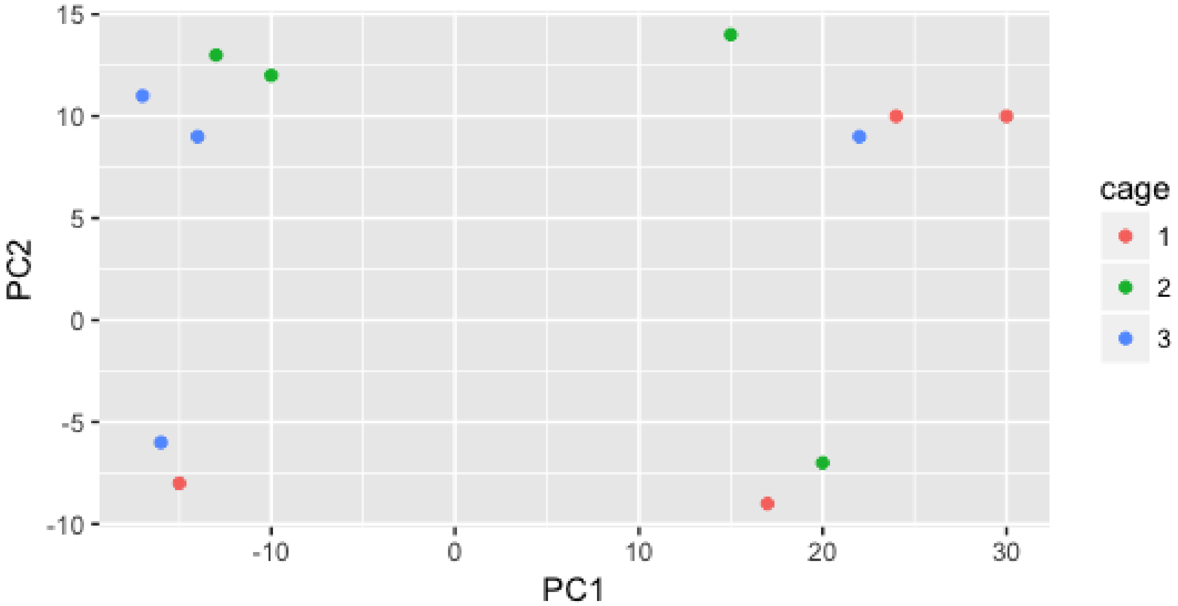
Week 6

- Differential Gene Expression Overview and Setup
- Count normalization
- Quality Control of DE Analyses
- DE Design Formulas



Courtesy of Paul Pavlidis, UBC

Week 6



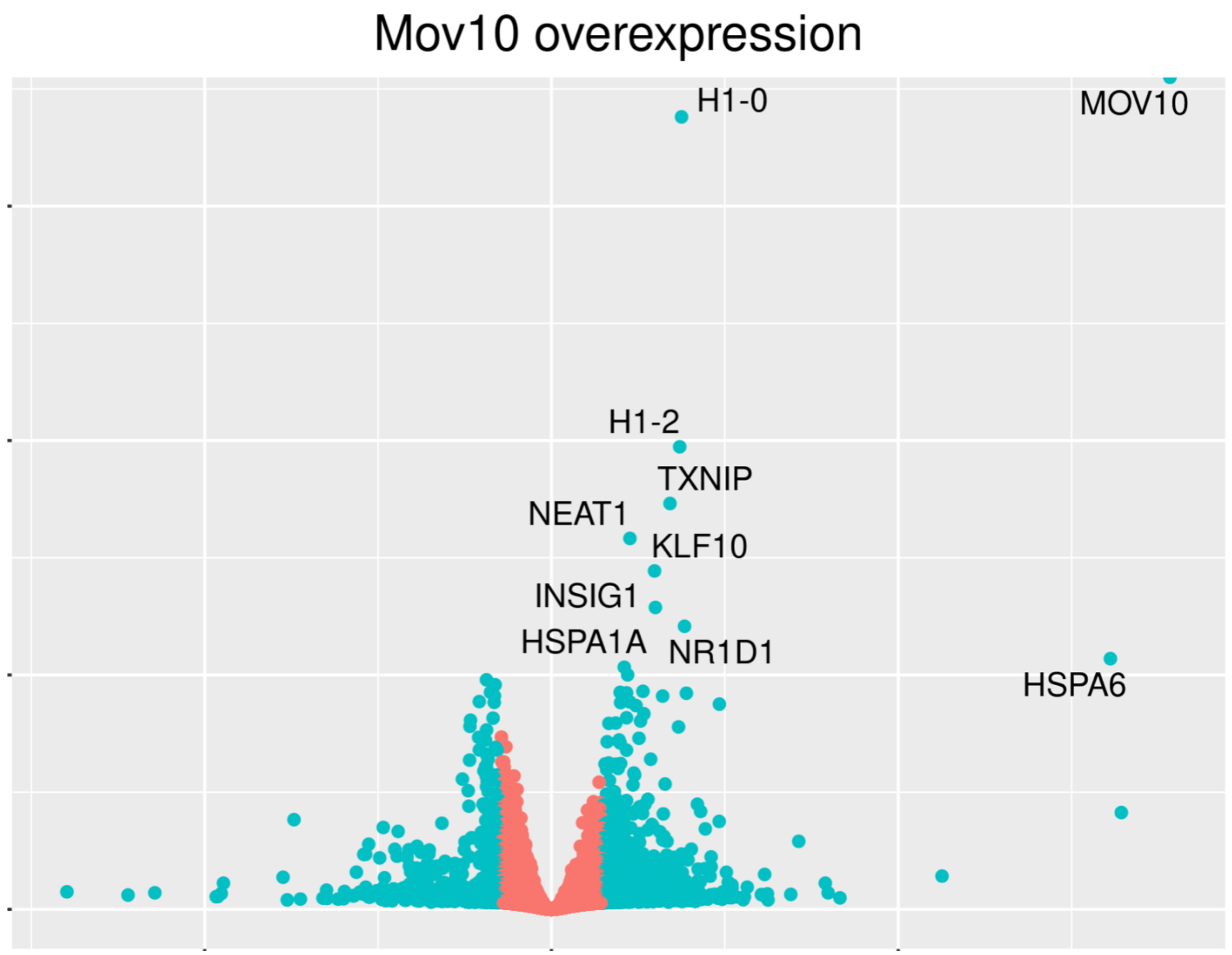
Does it look like cage explains a lot of the variation in these two PCs?

dds **<-** DESeqDataSetFromMatrix(counts, meta, design formula = ~sampletype)

Why do we normalize our count data before DE analysis?

Week 7/8

- Hypothesis Testing
- Exploring Wald Test Results
- Summarizing DE Results
- Visualizing DE Results



Week 7/8

What statistical distribution does DESeq2 use to model RNA-Seq count data?

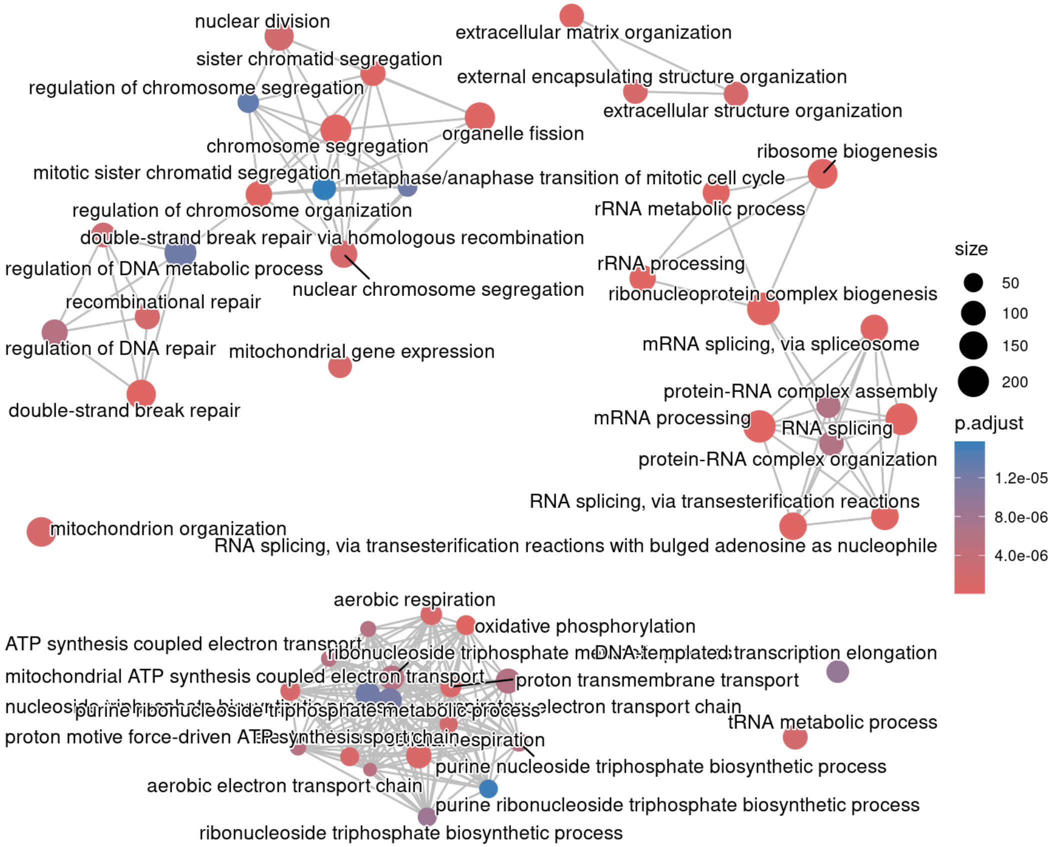
Fill in the blanks:

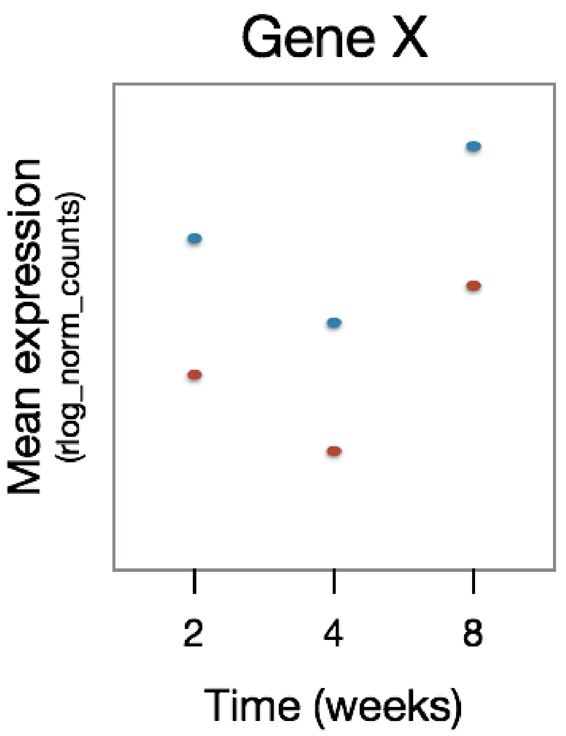
```
contrast_oe <- c("sampletype", mov10_oe , control )
```

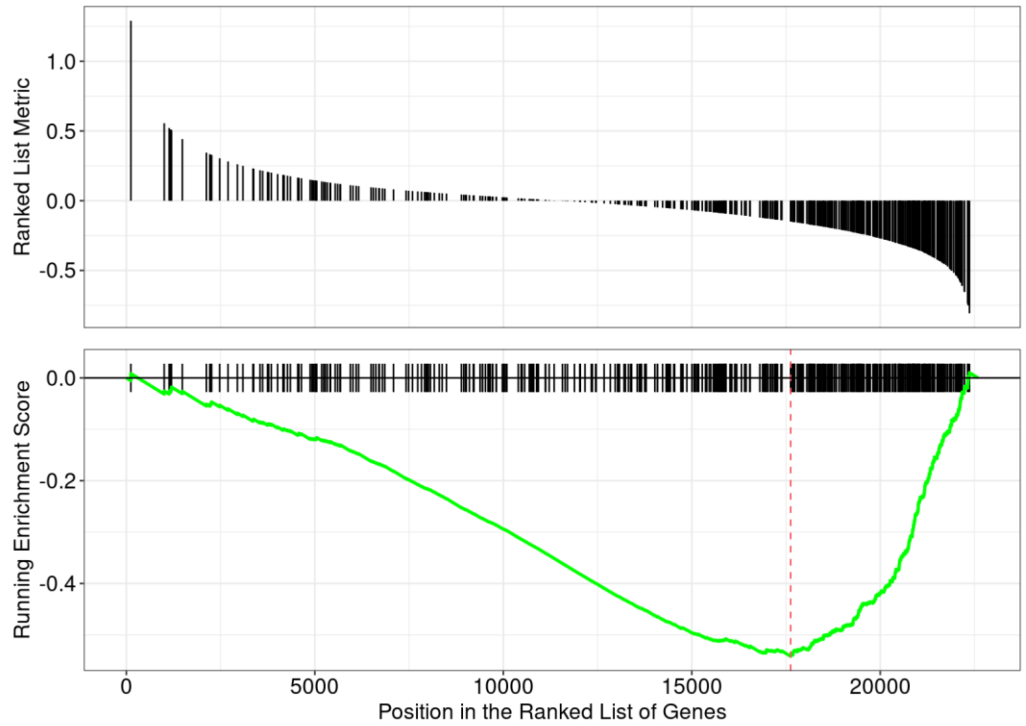
What is the difference between `pval` and `padj` in our DESeq2 results?

Week 8/9

- Likelihood Ratio Testing Results
- Annotation databases
- Overrepresentation analysis
- Functional class scoring







	ID	Description	GeneRatio	BgRatio	RichFactor	FoldEnrichment	zScore	pvalue	p.adjust	qvalue	geneID
GO:0080135	GO:0080135	regulation of cellular response to stress	309/6626	494/13585	0.6255061	1.282448	6.239957	2.514700e-10	0.0000014540	1.333850e-06	BAD/KDM1A/C
GO:0006417	GO:0006417	regulation of translation	223/6626	354/13585	0.6299435	1.291546	5.423482	3.481319e-08	0.0001006449	9.232825e-05	UPF1/YBX2/CS
GO:0008380	GO:0008380	RNA splicing	266/6626	435/13585	0.6114943	1.253720	5.248108	9.219386e-08	0.0001084644	9.950154e-05	UPF1/LUC7L/P
GO:0031329	GO:0031329	regulation of cellular catabolic process	261/6626	426/13585	0.6126761	1.256143	5.241338	9.568597e-08	0.0001084644	9.950154e-05	BAD/POLDIP2/
GO:0051052	GO:0051052	regulation of DNA metabolic process	284/6626	469/13585	0.6055437	1.241520	5.194035	1.234398e-07	0.0001084644	9.950154e-05	KDM1A/UPF1/
GO:0006282	GO:0006282	regulation of DNA repair	146/6626	221/13585	0.6606335	1.354468	5.184075	1.293814e-07	0.0001084644	9.950154e-05	KDM1A/CREBB
GO:0010506	GO:0010506	regulation of autophagy	218/6626	349/13585	0.6246418	1.280676	5.183293	1.313128e-07	0.0001084644	9.950154e-05	BAD/POLDIP2/
GO:0043484	GO:0043484	regulation of RNA splicing	118/6626	174/13585	0.6781609	1.390404	5.057365	2.528927e-07	0.0001827782	1.676745e-04	UPF1/PTBP1/C
GO:0006403	GO:0006403	RNA localization	126/6626	189/13585	0.6666667	1.366838	4.955454	4.364230e-07	0.0002803775	2.572088e-04	UPF1/MVP/ZC3
GO:0006397	GO:0006397	mRNA processing	277/6626	464/13585	0.5969828	1.223968	4.789909	1.005489e-06	0.0005813738	5.333326e-04	UPF1/PAF1/LU

Last items

- Please move your /data/Bspc-training/$USER directory to your main /data/$USER training folder.
  - Instructions to follow so we don't accidentally overwrite anything!
  - Paths you are using in scripts will change

- Course evaluation survey will go out this afternoon

- Office hours: April 23rd 10am-12pm

- Keep using course Slack channel – eventually move to general Help channel!